ML13 Features, targets and an honest baseline

Week 5, Baselines and a line | Session 13 of 24 | Machine Learning | New library: sklearn

Today's question. What did the session 1 model learn from, and how well would a plain guess do?

Objective. By the end: you can name the features and the target in a table, and calculate a baseline error that any model must beat.

Before this session: ML12 Checkpoint one and a project question. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
feature	An input known before you predict.	The clues in a puzzle.
training data	The examples the model learns from.	The worked examples in a textbook.
test data	Examples held back to check the model on things it has not seen.	The exam paper.
leakage	Using information that would not be known at prediction time.	Seeing the exam answers before the exam.
baseline	A simple guess a model must beat, such as the training mean.	Guessing that every home costs the average rent.

Block A. The baseline guess

A baseline uses no features at all. It guesses the average of the training targets every time.

Predict. Training prices are 2000, 4000 and 6000. Predict the baseline guess and its error when the real price is 5000. Write your guess on paper first.

Your guess, written before you run it

```
train_prices = [2000, 4000, 6000]
baseline = sum(train_prices) / len(train_prices)
actual = 5000
error = abs(actual - baseline)
print(baseline, error)
```

Run it, then compare. The cell prints 4000.0 and 1000.0. The training mean 4000.0 is the baseline guess. abs removes the minus sign, so the error is the distance between the guess and the real value, 1000 shillings. A model is worth using only if its errors on unseen examples are smaller than the baseline's.

Investigate.

1. Why must the baseline use only training prices?

2. What would the baseline guess be for every new price?

Modify. Change the actual price to 3000 and recalculate the error.

Block B. Open the session 1 model

The model from session 1 had one feature and one target. Print what it learned.

Predict. Name the feature and the target in this code before you run it. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
from sklearn.linear_model import LinearRegression
homes = pd.read_csv("synthetic_homes.csv")
X = homes[["area_m2"]]
y = homes["rent_ugx"]
model = LinearRegression().fit(X, y)
slope = model.coef_[0]
intercept = model.intercept_
print(round(slope), round(intercept))
```

Run it, then compare. The cell prints 2996 and 159436. The feature is area_m2, kept in X. The target is rent_ugx, kept in y. fit looked at all 120 homes and found a line: rent is about 159436 shillings plus 2996 shillings for each square metre. That model was scored on nothing at all. Judging it on the same homes it learned from would flatter it, so next session we keep some homes back.

Investigate.

1. Which of these is the feature and which is the target?
2. Why is it unfair to test the model on the same 120 homes?

Modify. Add rooms as a second feature and print the two slopes.

Make

For training values [2000, 4000, 6000], find the baseline guess for a new item. If its real value is 5000, calculate the absolute error. Name one leaked feature if you were predicting a selling price.

Save your work as ML13. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
NameError: name 'X' is not defined	The cell that creates X has not run yet.	Run the cells in order from the top.
KeyError: 'area'	The column is called area_m2 in this file.	Print homes.columns and copy the exact name.

Exit ticket

A model is useful only if its error on unseen examples is smaller than the baseline's error.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or assessments. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been submitted until you use the route in START_HERE.pdf.